GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering & Technology

Vulnerabilities in Gujarat and How to Overcome Them

Explanation of hazards, vulnerabilities and mitigation measures for the State of Gujarat

Student Name	_____	Enrollment No.	_____
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1. Aim

To explain the various **vulnerabilities in Gujarat** (natural, man-made, social and economic) and to describe how they can be **overcome** through structural, non-structural, institutional and community-based measures.

2. Introduction: Hazard, Vulnerability and Risk

A **hazard** is a natural or man-made event that can cause harm (for example an earthquake or a cyclone). **Vulnerability** is the condition of people, buildings and systems that makes them likely to be damaged by a hazard. **Capacity** is the strength and resources available to cope with it. A hazard becomes a **disaster** when it strikes a vulnerable community that has too little capacity. This is summarised as:

Risk = Hazard × Exposure × Vulnerability ÷ Capacity

Hazards cannot always be prevented, but **risk can be reduced by lowering vulnerability and raising capacity**. This is the idea behind all the measures suggested in this report. Vulnerability has several dimensions:

Type	Meaning	Example in Gujarat
Physical	Weak buildings and infrastructure in hazard-prone locations	Unreinforced masonry houses in Kutch
Social	Groups that are less able to protect themselves	Elderly, children, women, migrant workers, slum residents
Economic	Livelihoods that depend on the hazard-exposed environment	Fishing, salt-pan work, rain-fed farming, livestock rearing
Environmental	Degraded land, water and coast reducing natural protection	Loss of mangroves, over-extraction of groundwater
Institutional	Gaps in planning, enforcement and awareness	Unauthorised construction in flood plains

3. Gujarat at a Glance

- **Coastline:** about 1,600 km of India's 7,516 km coastline lies in Gujarat, with flat coastal terrain and a shallow continental shelf that favour storm surges. The Gulfs of Kutch and Khambhat can funnel and amplify surges.
- **Seismicity:** the Kutch region lies in Seismic Zone V, the highest category, and experiences some 200–300 small tremors in a month.

- **Climate:** large parts of Kutch, Saurashtra and North Gujarat are semi-arid with erratic rainfall, while South Gujarat receives heavy monsoon rain.
- **Rivers and industry:** rivers such as the Narmada, Tapi, Mahi and Sabarmati flow through densely populated, industrialised districts; chemical, petroleum and port industries are concentrated at places such as Jamnagar, Dahej, Ankleshwar, Vapi and Kandla/Mundra.
- **Hazard atlas:** the Gujarat State Disaster Management Authority (GSDMA) prepared a Hazard Risk and Vulnerability Atlas covering earthquakes, tsunamis, floods, cyclones, droughts and chemical/industrial hazards for every district and taluka.

Indicative hazard exposure by region

The table gives a simplified, indicative picture (H = high, M = medium, L = low) based on the general hazard profile of the State. District-level ratings are available in the GSDMA Hazard Risk and Vulnerability Atlas.

Region	Earthquake	Cyclone / tsunami	Flood	Drought	Heat wave	Industrial / chemical
Kutch	H	H	M	H	H	M
Saurashtra and coast	M	H	M	H	M	H
South Gujarat (Surat, Bharuch, Valsad)	M	M	H	L	M	H
Central (Ahmedabad, Vadodara, Anand)	M	L	H	L	H	H
North Gujarat	M	L	M	H	H	L

4. Vulnerabilities in Gujarat and Measures to Overcome Them

4.1 Earthquake

Why Gujarat is vulnerable: Kutch lies in Seismic Zone V, and a large part of the housing stock in towns and villages is built of unreinforced masonry or other non-engineered construction. Multi-storey buildings in fast-growing cities may not follow earthquake codes.

Past experience: The Bhuj (Kutch) earthquake of 26 January 2001 (magnitude about 7.7) killed around 20,000 people and destroyed towns such as Bhuj, Bhachau and Anjar. It led to the creation of the GSDMA.

How to overcome it:

- Enforce the earthquake-resistant design code for all new buildings and make technical inspection of designs and construction compulsory.
- Retrofit (strengthen) hospitals, schools, bridges and old buildings; identify and demolish unsafe structures.
- Train masons, engineers and architects in safe construction; promote earthquake-safe rural housing.
- Prepare micro-zonation maps of cities and use them in land-use planning.
- Conduct regular mock drills in schools, offices and neighbourhoods (‘Drop, Cover and Hold’); keep emergency kits ready.

- Strengthen search-and-rescue teams with equipment for collapsed structures; maintain seismological monitoring (Institute of Seismological Research).

4.2 Cyclones, Storm Surges and Tsunami

Why Gujarat is vulnerable: A long, flat coastline, a dense coastal population (including many coastal towns and ports) and fragile coastal housing. Fishing communities live and work close to the sea. Storm surges can penetrate far inland in creeks and gulfs. The coast is also exposed to tsunami risk from earthquakes in the Arabian Sea region.

Past experience: The 1998 Kandla/Porbandar cyclone was among the most destructive to strike the State. Cyclone Tauktae (May 2021) was the strongest to hit Gujarat since 1998, and Cyclone Biparjoy (June 2023) hit the Kutch–Saurashtra coast. Cyclone season in Gujarat is mainly May–June.

How to overcome it:

- Strengthen early-warning systems (IMD forecasts, SMS/alerts, loudspeakers) so that warnings reach the last fisherman and village.
- Evacuate people early from the surge zone to cyclone shelters; build and maintain multi-purpose cyclone shelters and approach roads.
- Build cyclone-resistant houses and infrastructure; protect power and communication lines.
- Conserve and plant mangroves and coastal shelter-belts; enforce Coastal Regulation Zone rules on construction.
- Provide boats with communication devices and ensure fishing boats return to harbour on warning.
- Prepare tsunami evacuation maps and signage; train coastal communities.

4.3 Floods

Why Gujarat is vulnerable: Heavy monsoon rainfall, dam releases, river-bed encroachment, unplanned urban growth and blocked drainage in cities. The Banaskantha area is prone to flash floods from sudden heavy rain, and South Gujarat river towns are exposed to releases from upstream dams.

Past experience: Surat was flooded in 2006 after large releases of water into the Tapi river; Banaskantha suffered severe floods in 2017; Vadodara faced flooding when the Vishwamitri river overflowed in 2024.

How to overcome it:

- Flood forecasting and warning, coordinated with dam operators; release water in a planned way and warn downstream areas in time.
- Keep rivers, flood plains and storm-water drains free of encroachment; desilt drains before the monsoon; apply flood-plain zoning.
- Build embankments, retention ponds and urban storm-water systems; raise critical facilities and houses in flood zones.
- Community flood plans: evacuation routes, relief camps, boats and life-saving equipment on standby.
- Promote crop insurance and flood-resistant livelihoods for affected farmers.

4.4 Drought and Water Scarcity

Why Gujarat is vulnerable: Kutch, Saurashtra and North Gujarat are semi-arid with variable rainfall. Groundwater has been heavily over-used for irrigation, leading to falling water tables and salinity ingress near the coast. Pastoral and small farming households depend on rain.

Past experience: The State has faced repeated droughts, including a severe drought around 1999–2002 that affected Kutch and Saurashtra.

How to overcome it:

- Water conservation: check dams, farm ponds, watershed development, rooftop rain-water harvesting and recharge of aquifers.
- Efficient irrigation (drip and sprinkler), suitable low-water crops and community water-management.
- Piped drinking-water supply and the linking of water sources to deficit areas; water tankers as a short-term measure.
- Fodder banks and cattle camps for livestock; drought insurance and employment programmes in lean periods.
- Drought monitoring with rainfall and satellite data to trigger early action.

4.5 Heat Waves

Why Gujarat is vulnerable: High summer temperatures, dense cities with concrete surfaces (urban heat island), and outdoor workers, slum residents, the elderly and children with little protection from heat.

Past experience: The 2010 heat wave in Ahmedabad was linked to more than a thousand excess deaths. Afterwards Ahmedabad introduced a Heat Action Plan (2013), an example that other Indian cities have followed.

How to overcome it:

- Heat early-warning system with colour-coded alerts and clear instructions for the public.
- Cool-roof programmes, tree planting and shaded public places; drinking-water points and cooling shelters.
- Adjust working hours for outdoor labourers; train hospitals to treat heat stroke.
- Public awareness: drink water, avoid peak sun hours, look after vulnerable neighbours.

4.6 Industrial, Chemical and Fire Hazards (Man-made)

Why Gujarat is vulnerable: Gujarat is a leading industrial State with many chemical, petrochemical and port facilities. Rapid growth, crowded buildings and weak enforcement of safety rules increase the risk of explosions, toxic releases, building fires and structural failures.

Past experience: The Surat coaching-centre fire (May 2019) killed 22 students, and the Morbi suspension-bridge collapse (October 2022) caused over 130 deaths, showing the cost of poor maintenance and crowd control.

How to overcome it:

- Strict safety audits and licensing of hazardous industries; on-site and off-site emergency plans with mock drills.
- Separate industrial zones from dense housing; keep buffer zones and emergency routes.
- Fire-safety compliance in schools, coaching centres, hospitals and malls; regular inspection of bridges and public structures.
- Crowd-management rules for fairs, bridges and religious gatherings; trained fire and hazmat teams.

4.7 Social and Economic Vulnerability

Why Gujarat is vulnerable: The same hazard harms people unequally. Fishing families, salt-pan workers in the Little Rann, pastoral communities, migrant labourers, slum dwellers, women, children, elderly and persons with disabilities have fewer resources to prepare, evacuate and recover.

Past experience: After the 2001 earthquake and the cyclones, poor and marginal families took the longest to recover their homes and livelihoods.

How to overcome it:

- Include vulnerable groups in planning: disability-friendly shelters, separate facilities for women, special care for the elderly and children.
- Livelihood protection: insurance, savings groups, alternative income and quick compensation after a disaster.
- Community-based disaster management: train local volunteers, women's groups and youth; local early-warning ‘champions’.
- Awareness and education: disaster management in school curricula, local-language campaigns, regular drills.

5. Summary: Vulnerability, Measures and Responsible Agencies

Vulnerability	Key structural measures	Key non-structural measures	Main agencies
Earthquake	Safe design, retrofitting, safe housing	Building codes, micro-zonation, drills	GSDMA, ISR, municipalities
Cyclone / tsunami	Shelters, strong coastal housing, mangroves	Early warning, evacuation, CRZ enforcement	IMD, GSDMA, district administration
Flood	Embankments, drains, retention ponds	Forecasting, zoning, dam-release protocol	Irrigation dept., municipalities, DDMA
Drought	Check dams, water supply, recharge	Crop planning, insurance, monitoring	Water-supply and agriculture depts.
Heat wave	Cool roofs, shelters, trees	Heat action plan, alerts, awareness	Municipal bodies, health dept.
Industrial / fire	Safe layout, fire equipment, bridge repair	Audits, licensing, emergency plans	Factories inspectorate, fire services
Social vulnerability	Inclusive shelters and housing	Training, livelihood support, education	GSDMA, NGOs, local bodies

6. Gujarat's Institutional Capacity

After the 2001 earthquake, Gujarat created the **Gujarat State Disaster Management Authority (GSDMA)**, the **Institute of Seismological Research (ISR)** and the **Gujarat Institute of Disaster Management (GIDM)** to plan, research and train. The State prepared a hazard atlas, runs earthquake and fire drills in schools, and works with district authorities under the **Disaster Management Act, 2005**. Early evacuation of coastal villages before recent cyclones, and city heat-action planning, show how preparedness can reduce the loss of life. The remaining challenge is to continue to enforce building and safety rules, to keep vulnerable groups in focus, and to adapt to the growing frequency of extreme weather linked to climate change.

7. Conclusion

Gujarat faces a wide range of hazards – earthquakes in Kutch, cyclones and storm surges on its long coast, floods in river and urban areas, droughts and heat in the semi-arid regions, and industrial and fire accidents. Disasters occur when these hazards meet physical, social and economic vulnerability. The vulnerability can be overcome by combining safe construction and protective works, effective early warning and planning, strong institutions, and the participation of well-informed communities. Reducing vulnerability before a disaster is far cheaper, in lives and money, than rebuilding afterwards.

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